

pdp11_70_memory_sop_v1.json

```
{
  "id": "SOP-PDP11-002",
  "title": "PDP-11/70 Memory Parity Error Isolation & Repair",
  "system": "PDP-11/70",
  "component": "MK11/MJ11 Memory System",
  "author": "SymbioStreams Semantic Bridge",
  "source": "Bitsavers (KA-K1170-MG-003)",
  "steps": [
    {
      "step": 1,
      "action": "Initial Error Identification",
      "details": "When the system halts or reports a parity error, use the console switches to examine the Memory System Error Register (MSER) at address 17 777 744.",
      "check": "Bit 15 should be set (1) indicating a parity error."
    },
    {
      "step": 2,
      "action": "Address Capture",
      "details": "Read the High Error Address Register (HEAR) at 17 777 740 and Low Error Address Register (LEAR) at 17 777 742 to find the physical address of the failure.",
      "target": "Record the 22-bit physical address."
    },
    {
      "step": 3,
      "action": "Run Memory Diagnostics",
      "details": "Boot XXDP and run the EKBB series diagnostics. Specifically, use EKBBE0 (Memory Management Part 1) and EKBBF0 (Memory Management Part 2).",
      "instruction": "Set Switch Register SW15=1 to halt on error for easier isolation."
    },
    {
      "step": 4,
      "action": "Board Level Isolation",
      "details": "If the diagnostic fails consistently, locate the memory board corresponding to the address captured in Step 2. Power down the system safely.",
      "action_item": "Swap the suspected memory board with a known good board from a different slot."
    },
    {
      "step": 5,
      "action": "Validation",
      "details": "Power on and re-run the EKBB diagnostics. Verify that the error has either disappeared or moved with the board.",
      "inference": "If the error moved, the board is defective. If it stayed in the same slot, the backplane or controller logic may be at fault."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "MSER Bit 15 is 0",
      "next": "No parity error detected. Check other system error registers."
    }
  ]
}
```

```
    {
      "condition": "MSER Bit 15 is 1",
      "next": "Read HEAR/LEAR for failing address."
    },
    {
      "condition": "Diagnostic passes after swap",
      "next": "Faulty board isolated. Replace board."
    }
  ]
}
```